Exercises for Genetic Algorithms

1. A simple Genetic Algorithm is designed to find the maximum value of

$$f(x) = \begin{cases} x, i \quad f0 \le x \le 32 \\ 64 - x, i \ f32 < x \le 63 \end{cases}$$

x=0,1,…,63 is encoded with a bitstring of length 6, f(x) is used as the fitness function, and single-point crossover and bitwise mutation are adopted for the genetic operations. The initial, randomly generated population is:

Chromosome label	Chromosome string
A	111010
B	011011
C	001001
D	000110

(1) Evaluate the fitness of each chromosome in the initial population, the overall fitness of the whole generation, and identify the best individual chromosome.
(2) After applying the selection operator, chromosomes A and B are chosen as parents to cross over after the third bit to produce chromosomes E and F. State the bitstrings E and F, and evaluate their fitness.
(3) Chromosomes B and C are selected to clone themselves into the next generation. Chromosome D is not selected and therefore becomes extinct. After crossover and cloning, offspring C is selected to mutate at the first bit to form Cm. State the bitstring Cm and evaluate its fitness
(4) State the new generation of chromosomes that results after the selection, crossover, clone and mutation steps performed in parts (i), (ii) and (iii). Evaluate the overall fitness of the generation and identify the best individual chromosome.

2. The Angle Sum and Difference Identities in trigonometry indicate that a trigonometric function of a sum or difference of two angles can be evaluated by simple arithmetic operations upon the trigonometric functions of the individual angles. For example

$$sin(x + y) = sin(x) \times cos(y) + cos(x) \times sin(y)$$

Describe how you would design a software program using genetic programming to discover these identities.

3. Delivery route scheduling

A postal worker needs to deliver N parcels to different addresses. Assuming the distances between these addresses and the branch are known, explain how you would design a genetic algorithm to find the shortest route for delivery. You need to specify:

(1) The encoding method;
(2) The fitness function;
(3) The genetic operators.

4. Delivery route scheduling with multiple carriers
For the same problem in the previous question, there are M postal workers in the branch and each of them is to be allocated N parcels to deliver (note the total number of parcels is M×N). How would you design a genetic algorithm to minimise the overall distance travelled by all the M workers? Again, you need to specify:
(1) The encoding method;
(2) The fitness function;
(3) The genetic operators.

5. Maintenance scheduling
There are seven power units in a power plant to be maintained in four equal intervals. The maximum loads expected during these intervals are 80, 90, 65 and 70MW. The capacities of the units and the duration of maintenance (in terms of number of continuous intervals) are shown in Table 1.2. Assume that:
(a) Maintenance starts at the beginning of an interval and continues through to the end of the same or an adjacent interval;
(b) The net reserve, which is defined by the total capacity of the units in use minus the maximum expected load, must be greater than or equal to zero at any interval.
Describe how you would design a genetic algorithm to solve this maintenance problem. You need to specify:
(4) The encoding method;
(5) The fitness function;
(6) The genetic operators.

Unit Number	Unit Capacity (MW)	Maintenance duration (Intervals)
1	20	2
2	15	2
3	35	1
4	40	1
5	15	1
6	15	1
7	10	1

Power units and their maintenance requirement.

6. An expression of propositions is in Conjunctive Normal Form (CNF) when it is a sequence of clauses joined by an AND (∧) relation, where each of the clauses is in the form of a disjunction, the OR (∨), of literals. For example, if the literals are *a, b, c, d, e* and *f*, then the following expression is in CNF.

$$(\neg a \vee c) \wedge (\neg a \vee c \vee \neg e) \wedge (\neg b \vee c \vee d \vee \neg e) \wedge (a \vee \neg b \vee c) \wedge (\neg e \vee f)$$

The CNF satisfaction problem is one where an assignment of *true* or *false* must be given to each of the literals in a CNF, so that the CNF evaluates to *true*. In the above example, the assignment 000000 (for *abcdef* respectively) is a solution.

Describe how you would design a genetic algorithm to solve the CNF satisfaction problem.